



Merit Education Group

What Is a Variable? — Answer Key

Algebra 1 • Unit 1 • Lesson 1 • Independent Practice

Standard: 9.3.5.7

Standard: 9.3.5.8

Tier 1: Foundational

Answers for vocab and identification (see §1.1–§1.2 in Guided Notes).

1. A **variable** is a symbol, usually a letter, that represents an unknown quantity.
2. In the expression $3x + 5$, the number 5 is called a **constant** because it does not change.
3. In $7y$, the number 7 is the **coefficient** and y is the **variable**.
4. Variables in $4a + 9 - 2b$: **a and b**
5. Coefficients in $12m + 3n - 8$: **12 and 3**

Tier 2: Application

Answers for translating word phrases (see §1.3 in Guided Notes).

6. A number increased by 9 **$x + 9$**
7. Seven less than a number **$x - 7$**

Note: “less than” reverses the order — $x - 7$, not $7 - x$.

8. The product of 5 and a number **$5x$**
9. Twice a number, decreased by 4 **$2x - 4$**

Tier 3: Extension

Unpack the problem, define the variable, write expressions, and show substitution (see §1.4 in Guided Notes).

10. Rectangular Garden

Given: w = width of garden.

Length: $L = 2w + 3$ (twice the width plus 3 more feet).

Perimeter: $P = 2L + 2w = 2(2w + 3) + 2w$

Simplify: $P = 4w + 6 + 2w = 6w + 6$

$$L = 2w + 3; P = 6w + 6$$

11. Cell Phone Plan

Given: t = number of text messages.

Monthly cost: $C = 25 + 0.10t$ (\$25 flat fee + \$0.10 per text).

For $t = 120$: $C = 25 + 0.10(120) = 25 + 12 = 37$

$$C = 25 + 0.10t; t = 120 \text{ gives } \$37$$